KENDRIYA VIDYALAYA SANGATHAN BHUBANESWAR REGION
FIRST PRE-BOARD EXAMINATION 2024-25
SUB-PHYSICS
CLASS – XII
SET 1

Maximum Marks: 70 Time Allowed: 3 hours

General Instructions

(1) There are 33 questions in all. All questions are compulsory.
(2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
(3) All the sections are compulsory.
(4) **Section A** contains **sixteen questions, twelve MCQ and four Assertion Reasoning based of 1 mark each**, **Section B** contains **five questions of two marks each**, **Section C** contains seven questions of three marks each, **Section D** contains **two case study-based questions of four marks each** and **Section E** contains **three long answer questions of five marks each**.
(5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.
(6) Use of calculators is not allowed.
(7) You may use the following values of physical constants wherever necessary

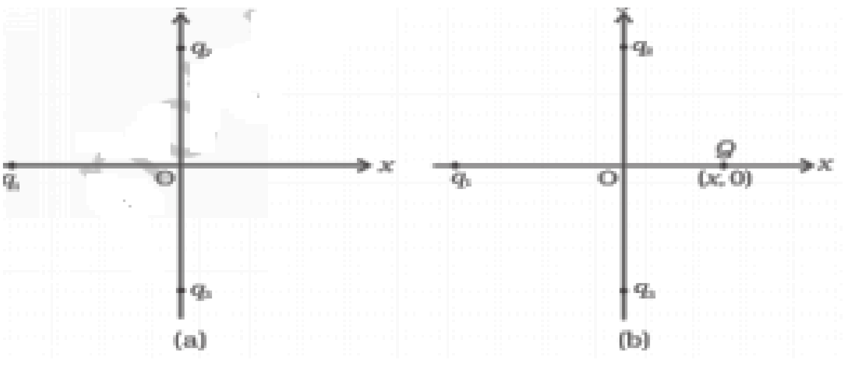
$e = 1.6 \times 10^{-19}$ C c = 3 x 108 m/s $m_e = 9.1 \times 10^{-31}$ kg

$m_p = 1.7 \times 10^{-27}$ kg $h = 6.63 \times 10^{-34}$ J s $\varepsilon_0 = 8.854 \times 10^{-12}\ C^2 N^{-1} m^{-2}$

$\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ N m² C^{-2} $\mu_o = 4\pi \times 10^{-7}$ T m A^{-1}

[SECTION-A]

Q1. In the given figure, two positive charges q2 and q3 fixed along the y axis, exert a net electric force in the + x direction on a charge q1 fixed along the x axis. If a positive charge Q is added at (x, 0), the force on q1



(A) shall increase along the positive x-axis.

(B) shall decrease along the positive x-axis.

(C) shall point along the negative x-axis.

(D) shall increase but the direction changes because of the intersection of Q with q2 and q3

Q2 Two uniformly charged spherical conductors A and B of radii 5 mm and 10 mm are separated by a distance of 2 cm. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitude of electric fields at the surface of the spheres A and B will be

(A) 1:2 (B) 2:1 (C) 1:1 (D) 1:4

Q3 In a parallel plate capacitor with air between the plates, each plate has an area of 6 X 10^{-3} $m^2$and the distance between the plates is 3 mm. The capacitance of the capacitor will be

(A) 16 pF (B) 18 pF (C) 20 pF (D) 0 pF

Q4 An electric current is passed through a circuit containing two wires of same material, connected in parallel. If the lengths and radii of the wires are in the ratio of 3:2 and 2:3, then the ratio of the current passing through the wires

(A) 2:3 (B) 3:2 (C) 8:27 (D) 27:8

Q5 Two wires of the same length are shaped into a square of side 'a' and a circle with radius 'r'. If they carry same current, the ratio of their magnetic moment is

(A) 2 : π (B) π : 2 (C) π :4 (D) 4 : π

Q6 In a series LCR circuit, the capacitance is changed from C to C/4. For the resonant frequency to remain unchanged, the inductance should be changed from L to nL, where n is

(A) $\frac{1}{2}$ (B) 2 (C) 4 (D) $\frac{1}{4}$

Q7. Correct match of column I with column II is

C-1 (waves)	C-II (wavelength range)
(1) Infra-red	P. > 0.1 m
(2) Radio	Q. 700 nm to 400 nm
(3) Light	R. 0.1 m to 1 mm
(4) Microwave	S. 1mm to 700 nm

(A) 1-P, 2-R, 3-S,4-Q (B) 1-S, 2-P, 3-Q, 4-R

(C) 1-Q,2-P, 3-S, 4-R (D) 1-S, 2-R, 3-P, 4-Q

Q8. In the wave picture of light, the intensity I of light is related to the amplitude A of the wave as

(A) I α $\sqrt{A}$ (B) I α A (C) I α A^2 (D) I α $\frac{1}{A^2}$

Q9. The coherence of two light sources means that the light waves emitted have

(A) same frequency (B) same intensity

(C) constant phase difference (D) same velocity

Q10. A small air bubble in a glass sphere of radius 2 cm appears to be 1 cm from the surface when looked at, along a diameter. If the refractive index of glass is 1.5, then the true position of the air bubble will be

(A) -1.5 cm (B) -2.0 cm (C) 1.4 cm (D) -1.2 cm

Q11. The ground state energy of hydrogen atom is -13.6 eV. The potential energy of an electron in the second excited state is

(A) -3.02 eV (B) -1.70 eV (C) -0.85 eV (D) -1.51 eV

Q12. At equilibrium, in a p-n junction diode the net current is

(A) due to diffusion of majority charge carriers.

(B) due to drift of minority charge carriers.

(C) zero as diffusion and drift currents are equal and opposite.

(D) zero as no charge carriers cross the junction.

For Questions 13 to 16, two statements are given –one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. If both Assertion and Reason are true and Reason is the correct explanation of Assertion.

B. If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

C. If Assertion is true but Reason is false.

D. If both Assertion and Reason are false.

Q13. **Assertion (A):** To increase the range of an ammeter**,** a high resistance in series must be connected to it.

Reason (R): The ammeter with increased range should have high resistance.

Q14. **Assertion (A):** de-Broglie waves are not electromagnetic waves.

Reason (R): Electromagnetic waves are associated with accelerated charge particles.

Q15. **Assertion (A):** Angular momentum of single electron in any orbit of hydrogen type atoms is independent of the atomic number of the element

Reason (R): In ground state angular momentum is minimum.

Q16 **Assertion (A) :** Binding energy per nucleon is nearly constant for middle mass nuclei ($30 < A < 170$).

Reason (R): Nuclear force is short ranged in nature.

[SECTION – B]

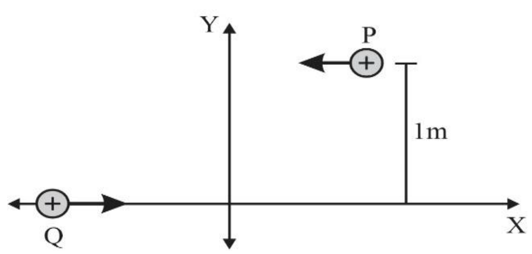
Q17. Draw a schematic arrangement of the Geiger-Marsden experiment. What is impact parameter? How it is related with angle of scattering of α-particles.

Q18. (I) In a Young's double slit experiment, fringes are obtained on a screen placed a certain distance away from the slits. If the screen is moved by 5 cm towards the slits, the fringe width changes by 30 μm. Given that the slits are 1 mm apart, calculate the wavelength of the light used.

OR

(II) In Young's double slit experiment using monochromatic light of wavelength λ, the intensity of light at a point where path difference is $\lambda/3$, is K units. What is intensity at the centre of the bright fringe?

Q19. P and Q are two identical charged particles each of mass 4×10^{-26} kg and charge 4.8×10^{-19} C, each moving with the same speed of 2.4×10^{5} m/s as shown in the figure. The two particles are equidistant (0.5 m) from the vertical Y -axis. At some instant, a magnetic field B is switched on so that the two particles undergo head-on collision.



Find –

(I) the direction of the magnetic field and

(II) the magnitude of the magnetic field applied in the region.

Q20. Draw a graph showing the variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei (A > 170) decreases with the increase in mass number. Explain.

Q21. A potential difference V is applied to a conductor of length l, diameter D. How are the electric field E, the drift velocity v_d and the resistance R affected when (i) V is doubled (ii) l is doubled.

[SECTION – C]

Q22. Draw the circuit diagrams for obtaining the V-I characteristics of a p-n junction diode. Explain briefly the salient features of the V-I characteristics in (i) forward biasing, and (ii) reverse biasing.

Q23. A parallel plate capacitor is charged by a battery. After sometime the battery is disconnected and a dielectric slab of dielectric constant K with its thickness equal to the plate separation is inserted between the plates. How it will affect

(i) the capacitances of the capacitor

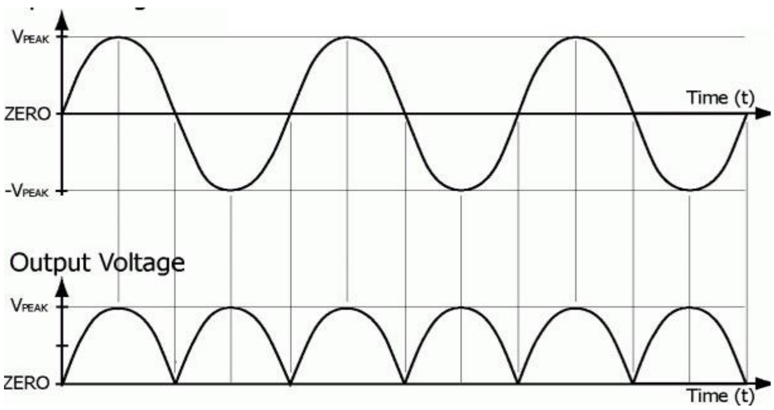
(ii) potential difference between the plates

(iii) the energy stored in the capacitors.

Justify your answer in each case.

Q24. Draw a labelled ray diagram showing the formation of an image by an astronomical refracting telescope in normal adjustment. Hence obtain the expression for its magnifying power.

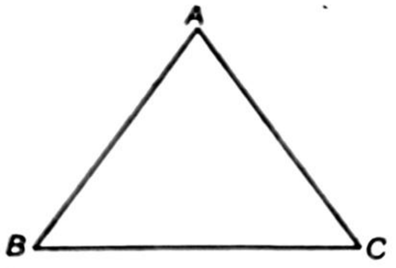
Q25. A sinusoidal input voltage applied to a device (X) produces an output voltage as shown in the graph given below. Name the device (X) and explain its working with the help of circuit diagram.



Q26. On a smooth plane inclined at 30° with the horizontal, a thin current-carrying metallic rod is placed parallel to the horizontal ground. The plane is located in a uniform magnetic field of 0.15 T in the vertical direction. For what value of current can the rod remain stationary? The mass per unit length of the rod is 0.30 kgm^{-1}.

Q27. (i) A ray of light incident on face AB of an equilateral glass prism, shows minimum deviation of 30°. calculate the speed of light through the prism.

(ii) Find the angle of incidence on face AB, so that the emergent ray grazes along the face AC.



Q28. (I) State Gauss's law in electrostatics. Using this law, derive an expression for the electric field due to a uniformly charged infinite plane sheet.

OR

(ii) (a) Define electric flux and write its SI unit.

(b) Use Gauss's law to obtain the expression for the electric field due to an infinitely long straight wire of linear charge densityλ.

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[SECTION – D]

Q29. Case Study Based Question

Ferromagnetic substances are those which are strongly magnetised by relatively weak magnetic field and in the same sense as the applied magnetic field. Iron nickel, cobalt, gadolinium and their alloys are ferromagnetic. These substances have relative permeability of the order of hundreds and thousands. The flux density B in a ferromagnetic substance is not directly proportional to the magnetizing force H. Hence the permeability ($\mu = \frac{B}{H}$) is not a constant. The permeability decreases with rise in temperature. Above curie temperature, a ferromagnetic substance becomes a paramagnetic substance.

(i) Which of the following is incorrect statement?

(a) Magnetic intensity is a vector quantity.

(b) Induced magnetization is a process where you can magnetise a magnetic material placing near magnet.

(c) Magnetic intensity and intensity of manetisation are the same.

(d) None of these.

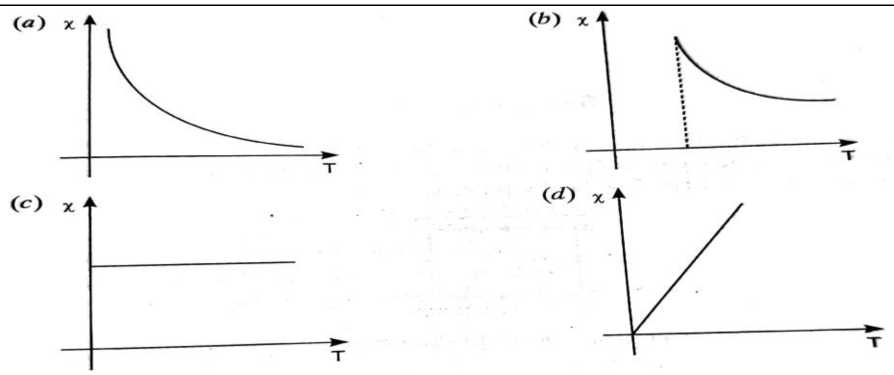
(ii) SI unit of magnetic flux is

(a) Ohm (b) Weber (c) Tesla (d) Weber/m^2

(iii) Which of the following has higher magnetic susceptibility?

(a) Diamagnetic (b) paramagnetic (c) Ferromagnetic (d) None of these

(iv) The magnetic susceptibility χ of a ferromagnetic material varies with temperature (T) as



OR

(V) There are four light-weight rod samples A,B,C,D separately suspended by threads. A bar magnet is slowly brought near each sample and the following observations are noted

(I) A is feebly repelled. (II) B is feebly attracted.

(III) C is strongly attracted (IV) D remains unaffected.

Which of the following is correct?

(a) C is of diamagnetic material. (b) D is of a ferromagnetic material.

(c) A is of a non-magnetic material. (d) B is of a paramagnetic material.

Q30. Case Study Based Question

The salient features of photoelectric effect are enumerated below.

An electromagnetic wave travels in the form of discrete packets or bundles of energy called quanta or photon. All photons emitted by any source travel through free space carrying an energy $E=h\nu$ and momentum $p = h/\lambda$. The photon energy depends on the frequency ν of the radiation and not its intensity. The transfer of discrete amount of energy from a photon to a free electron inside the photosensitive metal is the basic cause of photoelectric emission.

(i) Above the threshold frequency the maximum kinetic energy of the emitted photo-electrons increases linearly with

(a) intensity of incident radiation (b) frequency of incident radiation

(c) apparent time of the incident radiation (d) none of these

(ii) Light of wavelength 200nm falls on a metal surface of work function 4.2 eV. The maximum kinetic energy of the electrons emitted is about

(a) 6.2 eV (b) 2.0 eV (c) 4.2 eV (d) 4.0 eV

(iii) The threshold frequency for photoelectric effect on a metal corresponds to a wavelength of 300 nm. Its work function is about

(a) 4.0×10^{-19} J (b) 1 J (c) 6.6×10^{-19} J (d) 3.3×10^{-19} J

(iv) The stopping potential for a photoelectric emission process is 2V. Find the maximum kinetic energy of the electrons ejected in the process.

(a) 3.2×10^{-18} J (b) 1.6×10^{-19} J (c) 3.2×10^{-19} J (d) zero

OR

(v) For which of the following radiations, the cut-off voltage will be maximum

(a) violet (b) ultraviolet (c) X rays (d) infrared

[SECTION – E]

Q31. (I) a) Define resistivity. Explain why resistivity of metals increases and that of semiconductors decreases with rise in temperature.

b) A series battery of 6 lead accumulators each of emf 2.0 V and internal resistance 0.50 Ω is charged by a 100 V d.c. supply. What series resistance should be used in the charging circuit in order to limit the current to 8.0 A? Using the required resistor , obtain the power supplied by the d.c. source.

OR

(II) a) Using Kirchhoff's laws obtain the equation of the balanced state in a Wheatstone bridge.

b) Two cells of emfs 1.5 V and 2 V and internal resistances 2 Ω and 1 Ω respectively have their negative terminals joined by a wire of 6 Ω resistance and positive terminals by a wire of 4 Ω resistance . A third resistance wire of 8 Ω connects middle points of these wires. Draw the circuit diagram. Using Kirchhoff laws, find the potential difference at the end of this third wire.

Q32. (I) Explain briefly, with the help of a labelled diagram, the basic principle of the working of an a.c. generator.

In an a.c. generator, coil of N turns and area A is rotated at an angular velocity ω in a uniform magnetic field B. Derive an expression for the instantaneous value of the emf induced in coil.

What is the average value of emf in one complete revolution of the coil?

What is the source of energy generation in this device?

OR

(II) a) With the help of a diagram, explain the principle of a device which changes a low ac voltage into a high voltage . Deduce the expression for the ratio of secondary voltage to the primary voltage in terms of the ratio of the number of turns of primary and secondary winding. For an ideal transformer, obtain the ratio of primary and secondary currents in terms of the ratio of the voltages in the secondary and primary coils.

b) Write any two sources of the energy losses which occur in actual transformers.

c) A step-up transformer converts a low input voltage into a high output voltage. Does it violate law of conservation of energy? Explain.

Q33 (I) a) Draw a ray diagram showing the formation of a real image of an object placed at a distance ‘u’ in front of a concave mirror of radius of curvature ‘R’. Hence, obtain the relation for the image distance ‘v’ in terms of u and R.

b) A 1.8m tall person stands in front of a convex lens of focal length 1 m, at a distance of 5m. Find the position and height of the image formed.

OR

(II) a) Draw a ray diagram showing refraction of a ray of light through a triangular glass prism. Hence obtain the relation for the refractive index (n) in terms of angle of prism (A) and angle of minimum deviation (δ_m)

b) The radii of curvature of the two surfaces of a concave lens are 20 cm each. Find the refractive index of the material of the lens if its power is -5.0 D.